

USING DIFFUSION MODELS TO FORECAST ELECTRIC VEHICLE OUTLOOK IN FINLAND, SWEDEN, AND NORWAY

A forecast of electric vehicle adoption

Minh Khoa Pham

International Business

Bachelor's Thesis

Supervisor: Christopher Decker

Date of submission: 17 March 2023

Aalto University

School of Business

Bachelor's Program in International Business

Mikkeli Campus

Author: Minh Khoa Pham Title of thesis: Using diffusion models to forecast electric vehicle outlook in Finland, Sweden, and Norway
Date: 17 March 2023
Degree: Bachelor of Science in Economics and Business Administration
Supervisor: Christopher Decker
Objectives: This research aims to compare the forecasting performance of the Bass, Logistic, Gompertz, and Generalized Bass models. The study also analyzes the impact of charging points on electric vehicle (EV) adoption rates in Finland, Sweden, and Norway. Finally, the research investigates the effects of the COVID-19 pandemic on EV markets in the three Nordic countries. This paper will provide insights into the EV diffusion process and policy incentives promoting sustainable transportation. Summary: This research predicts EV adoption in Finland, Sweden, and Norway until 2030 using four diffusion models. Parameters are estimated using the GRG Nonlinear method and assume a 30% market potential as per the EV30@30 initiative. The best-fit models are selected based on error measures. Conclusion: The findings show variations in the diffusion rate across Finland, Sweden, and Norway, and the Gompertz and Logistic models perform the best in forecasting EV diffusion in the three countries. The results suggest the three countries' EV sales can reach 30% market share by 2030. The study also examines the relationship between the number of charging points and EVs. Moreover, market uncertainty like the COVID-19 pandemic also affected electric car and electric bus uptakes.
Keywords: diffusion models, Bass, Gompertz, logistics, generalized bass model, forecast, electric car, electric bus, Finland, Sweden, Norway Language: English
Grade:

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1. INTRODUCTION

1.1. Background

There are growing concerns among governments and policymakers about the increase in greenhouse gas and carbon emission. The road transportation sector significantly contributes to this problem accounting for 32 percent of global CO₂ emissions (IEA, 2021). According to Requia et al. (2018), electric vehicles (EVs) - vehicles using electric motors to operate, are viable alternatives since they produce fewer pollutants and require lower fuel prices than traditional internal combustion engine (ICE) vehicles. EVs can also help reduce atmospheric emissions and negative impacts on human health. Thus, several European countries have implemented the electric mobility program, transitioning the vehicle fleet from ICE to electric. As a result, in Europe, the share of electric vehicle sales has increased from 1 percent in 2016 to 17 percent in 2021. Due to its great capability to decrease greenhouse gas emissions worldwide, policymakers must devise a plan for electric vehicle (EV) adoption, as it is connected to future electricity demand and the electrical grid capacity. Forecasting future demand for EVs in private and public sectors would help governments and policymakers to provide the proper support to adopt the innovation. Forecasting the demand for electric vehicles (EVs) is difficult because the technology is relatively new to different markets. As a result, traditional time-series models may not be appropriate for predicting the demand for new products such as EVs. In addition, many current studies on predicting EV demand rely on surveys of stated preferences and do not consider actual sales data (Kumar et al., 2021). This can lead to inaccurate forecasts and make these studies subject to error.

1.2. Research Problem

To evaluate EV adoptions in the market, it is important to understand how EV technology tends to spread. To achieve this, diffusion rate and time series methods can be employed to forecast and analyze the diffusion process. As a result, this study solely concentrates on utilizing diffusion rate and time series methods to comprehend how EVs diffuse. In addition, although public electric vehicles, such as electric buses, have much potential to meet environmental goals and transportation demand for developed markets, no research

has been conducted to study and forecast the electric bus market. Thus, this paper will analyze electric passenger car and electric bus diffusion.

Moreover, there is a lack of literature on forecasting EV diffusion in developed markets. Nordic countries, notably Norway, have some of the highest EV adoption rates in the world. By studying and forecasting EV demand in the Nordics, this thesis can provide a deeper understanding of the factors that drive EV adoption and apply those insights to other regions. Furthermore, the findings can help governments, policymakers, car manufacturers, and electricity utilities develop the best action to adopt the innovations. Thus, this paper will study the markets and forecast demand for EVs in three Nordic countries: Finland, Sweden, and Norway.

1.3. Research Questions

This research aims to answer the following questions:

1. Which models would best forecast the electric car and electric bus demands in Finland, Sweden, and Norway?
2. How do charging points affect EV diffusion in Finland, Sweden, and Norway
3. What are the effects of the COVID-19 pandemic on electric car and electric bus diffusions in Finland, Sweden, and Norway?

1.4. Research Objectives

The research objectives are as follows:

1. To compare the performance of the Bass diffusion, Logistic growth, Gompertz curve, and Generalized Bass model in forecasting electric vehicle diffusion rates.
2. To analyze the influence of charging points on EV adoption in Finland, Sweden, and Norway
3. To study the effects of a pandemic on EV adoption in the three countries.

1.5. Research Structure

This research begins with reviewing the literature on previous studies about EV adoption barriers in the Nordics and different approaches to comprehend the technology adoption.

An overview of diffusion studies and four different diffusion models: the Bass model, Logistic model, Gompertz model, and Generalized Bass model is also included. Then, the paper uses historical data to find the models' parameters. Afterward, the results of each model will be compared and assessed. Moreover, government policies and future targets for EV adoption are analyzed. Finally, findings and recommendations are provided for policymakers and future researchers.

2. LITERATURE REVIEW

2.1. Introduction

The transition to electric mobility in Nordic countries has been ongoing for the past ten years (RenaultGroup, 2019). The governments introduced incentives and initiatives to increase electric vehicle (EV) adoption in the private and public sectors. Therefore, forecasting future demand for EVs in private and public sectors would help governments and policymakers to provide the proper support to adopt the innovation. This literature review discusses the studies on the EV market and related policies in three Nordic countries: Finland, Sweden, and Norway. Furthermore, it examines the use of diffusion models to forecast technology adoption. Previous studies have applied the models for EV forecasting in different countries with varying results. In addition, each model is discussed in detail.

2.2. Electric Vehicle Adoption

Previous research has focused on identifying and understanding EV adoption factors. Noel et al. (2020) aimed to characterize the barriers to electrification, the barriers' perceived futures, and policy implications in the Nordic region. To gain this insight, the researchers interviewed transportation and electricity experts in Finland, Sweden, and Norway. They found that EVs still face many barriers even in developed economies like the Nordic region. Specifically, EVs' price, range, and charging are the main barriers to consumer adoption. However, since Noel et al. (2020) analyzed the results based on only expert opinions, the results may not represent the actual views of customers.

On the other hand, more studies have surveyed customers to provide more relevant results about adoption behavior. Based on consumer survey data, Vassileva & Campillo (2017) examined the positive features of Sweden for large-scale EVs diffusion. The researchers also looked at the impediments to EV penetration. Then they identified the typical demographics of an EV owner in Sweden. They discovered that the typical EV owner is male with a medium-high income and a high level of education.

Likewise, Sovacoo et al. (2018) studied how population demographics impact electric mobility preferences in the Nordics. They found that Norway and Finland also have the same typical demographics of an EV owner as Sweden, males with medium-high incomes and high levels of education. As a result, the researchers concluded that policies targeting a general audience might not be as effective as those targeting different demographics (Sovacool et al., 2018).

These studies aimed to reveal EV market segments, understand future EV adoption, and provide insights into transport and energy policy with a focus on sustainability. In general, the research indicates that demographic factors, government policies, and access to charging stations are crucial in promoting the uptake of electric vehicles in the Nordic region. However, they also find that the high initial cost and concerns about the distance one can travel on a single charge are significant obstacles to broader adoption.

Overall, there are many methods to understand EV adoption. The forecasting method in this study will be used as another way to study the EV market, contributing to the existing literature. Forecasting the EV markets can help policymakers better understand their countries' potential and develop the necessary instruments.

2.2.1. Finland

There has been a significant increase in demand for EVs in Finland due to technological improvements and a greater focus on reducing greenhouse gas emissions, which contribute to climate change (IEA (International Energy Agency), 2022). According to EAFO (European Alternative Fuels Observatory) (2022), Finland's number of electric passenger vehicles has gradually increased from 614 in 2015 to 42,985 in 2022. Moreover, Finland's number of electric buses rose from

15 in 2016 to 424 in 2022. This growth can be attributed to various factors, including government policies and incentives to promote EV adoption and increase consumer interest in more sustainable and environmentally friendly transportation options (Helsinginilmastoteot, 2022).

First, the Finnish government is committed to reducing greenhouse gas emissions and promoting renewable energy use (ym.fi). To achieve these goals, the Finnish government has implemented various policies and incentives designed to encourage the adoption of EVs, including tax breaks and subsidies for electric vehicle purchases, as well as the construction of charging infrastructure. Second, the EV market in Finland has also been supported by several private-sector initiatives, including developing charging networks and services provided by energy companies such as Neste and Helen (helsinkitimes.fi). Third, Finland is home to several EV manufacturers and suppliers, and the government provides funding for research and development in the EV industry to support the sector's growth. These efforts have helped to increase the availability and accessibility of EVs for Finnish consumers, further driving their adoption in recent years.

Nevertheless, according to Yle (2022), although fully electric cars in Finland more than doubled in 2022, they still only accounted for 1.2% of the country's passenger vehicles, a much lower percentage than in Norway and Sweden. This might be due to Finland's initial focus on biofuels as an alternative fuel for vehicles and a lack of pioneering programs to promote EVs from 2017, when Norway and Sweden were already world leaders (Yle, 2017). In addition, there was a lack of charging stations in Finland. Currently, the government aims to have a total of 700,000 electric vehicles in Finland by 2030 (Yle, 2022). Overall, while the Finnish EV market is still at the beginning of electric vehicle diffusion compared to Norway and Sweden, the market has much potential for growth.

2.2.2. Sweden

Sweden is one of the leaders in using and manufacturing EVs (EY, 2022). The electric vehicle market in Sweden has been growing in recent years. The growth is

driven partly by the country's robust energy ecosystem and high consumer uptake. As stated by the European Alternative Fuels Observatory (2022), there were 7,532 electric passenger cars in Sweden in 2016 and 173,595 in 2022. Moreover, the number of electric buses went from 38 in 2016 to 818 in 2022.

The Swedish government introduced several policies in the form of subsidies and pricing to promote EV adoption (unfcc.int). These policies include financial incentives such as grants and tax breaks for electric vehicle buyers and infrastructure development initiatives such as installing charging stations. According to Egnér and Trosvik (2018), an increased number of public charging points, especially in urban cities, positively affects the adoption rate in Sweden. As a result, in 2022, the total EV and plug-in hybrid electric vehicle (PHEV) market share exceeded 50 percent (EAFO, 2022).

Moreover, the private industry has played a significant role in developing electric vehicles in Sweden. Technological advancements have decreased EV battery costs, making EVs more affordable. Additionally, according to the European Investment Bank (EIB, 2022), private organizations have invested in building charging stations and have started incorporating EVs into their fleet of vehicles, thereby boosting the accessibility and availability of EVs for customers.

However, on 8 November 2022, Sweden's new government removed EV purchase subsidies for vehicles ordered after 8 November 2022 (electrive, 2023). Their justification for no longer providing a state subsidy for electric cars is that EV's market shares, purchase prices, and operating costs are now similar to traditional gasoline or diesel cars. Therefore, the EV diffusion rate in Sweden can be expected to slow down.

On the other hand, Sweden has declared that it will ban internal combustion engine (ICE) car sales after 2030 (electrive, 2021). The ban is aligned with the government's climate goals of achieving zero net greenhouse gas emissions after 2045 (naturvardsverket.se).

2.2.3. Norway

Norway has been one of the leading countries in adopting and diffusing electric vehicles (EVs), with a relatively high market share of 86% compared to other countries (IEA, 2022). According to EAFO (2022), the number of electric cars in Norway went from 97,615 in 2016 to 512,979 in 2022. The number of electric buses increased from 0 in 2016 to 870 in 2022. The growth in Norway's EV market is largely due to a range of government electromobility local and national policies promoting EV use and supporting the development of the necessary infrastructure (Mersky et al., 2016).

Mersky et al. (2016) reported that tax pricing incentives, including tax exemptions and reductions for EVs, are among Norway's best policies to encourage consumers to adopt EVs. Additionally, the Norwegian government introduced other policies, including expanding charging stations, in the form of grants and through partnerships with private companies. Moreover, services and companies in Norway have also begun changing their vehicle fleets to electric ones (TØI, n.d.). These changes have helped further to increase the availability and accessibility of EVs to consumers.

Norway has successfully promoted EV diffusion through government policies and private-sector initiatives. These efforts have led to a significant increase in EVs on the road and have helped establish Norway as a leader in EV adoption and diffusion. However, more can be done, as the current market share of EVs in Norway is relatively high, but the country is yet to reach the goals set out in its national strategy for the transition to electric transport. The Norwegian Parliament set a goal for all new cars sold to be zero-emission by 2025 and for the entire vehicle fleet to be fossil-free by 2050 (elbil.no). Therefore, this paper will forecast the Norwegian EV market to determine if the country can reach its goals.

2.3. Diffusion Studies

Diffusion of innovation is a theory seeking to understand the process of adopting new technology in a population (Everett, 2003). Although the diffusion of innovation theory has

been used in many articles across different disciplines, it has a few drawbacks. Since the theory only focuses on the adopters deciding to adopt the technology, it does not consider exogenous variables, such as pricing and policies (Kemp & Volpi., 2008). The theory also considers innovation adoption independent, even though innovation adoption can increase due to another technology development.

2.3.1. Forecasting Technology Diffusion

Several methods are used to forecast the spread of EV markets. First, the consumer choice method combines the logit and individual models to comprehend the individual consumer decision (Soumia & Maaroufi, 2018). Gnann et al. (2018) determined four traits: (1) traits that impact the buying choices, (2) qualities of vehicles taken into account, (3) characteristics that depict consumer behavior, and (4) factors related to plug-in EVs to understand market diffusion. Because these factors are vastly different in each country, comparing the model results is challenging (Gnann et al., 2018).

Another forecasting method is using diffusion models. Diffusion models are a popular method for forecasting the acceptance and dissemination of new products, technologies, and developments. According to Bass et al. (1990), the approach behind these models is that the process of accepting a new item or technology follows a predictable pattern, known as the "diffusion" process, that can be modeled and forecasted.

There have been some studies applying diffusion models to their forecast. For example, Qian and Soopramanien (2014) used Gompertz, Logistic, and Bass models to forecast car sales in emerging Chinese markets using car ownership data. Furthermore, Soumia and Maaroufi (2018) used Bass, Logistic, and Gompertz models to predict the electric vehicle market in Morocco. Kumar et al. (2022) also used the Bass model to forecast market diffusion in countries with the most advanced EV technology.

These studies have shown that it is possible to apply diffusion models to EV technology in different countries. Additionally, this is the most commonly used

method to predict the diffusion of innovations. Therefore, this method has been chosen to forecast EV sales in the three Nordic countries. Three diffusion models, Bass, Gompertz, and Logistic, have been chosen to forecast the EV. The Generalized Bass model will also be used to address the Bass model's limitations.

2.3.2. Bass Model

The Bass Diffusion Model is a mathematical model that predicts the spread and adoption of new products or technologies. It was developed by Frank Bass in 1969 and is widely used in marketing and innovation research (Bass, 2004). It is built upon the idea that the likelihood of new innovation adoptions at a given time is related to the number of previous adoptions. The model can predict the adoption of a wide range of new products and technologies, including consumer goods, medical treatments, and even new technologies like electric vehicles (Kumar et al., 2022).

According to Bass (2004), the model assumes that a new product or technology adoption process follows an S-shaped curve, with an initial slow period of adoption, followed by a period of rapid adoption and eventually leveling off to a steady state. The model includes parameters, such as innovation and imitation factors driving the adoption of new products or technologies. The innovators are the initial adopters, while the imitators adopt later. In general, the Bass model uses the following equation to show the percentage of the population that adopted the technology at time t :

$$\frac{dF(t)}{dt} = [p + qF(t)][1 - F(t)] \quad (1)$$

Where $F(t)$ represents the proportion of the population who have adopted the technology up until time t using the cumulative distribution function. Moreover, p is the coefficient of innovation, and q is the coefficient of imitation. It is worth noting that 'innovators' are not influenced by the later adoption phases, while 'imitators' are influenced by the adoption decisions of innovators. In the beginning, p is the only factor determining adoption. Later, as the technology begins to be diffused, q

becomes more important. Finally, the adoption rate slows when $F(t)$ reaches 1. In addition, let K be the market potential and $F(t)$ be the cumulative adoption at time t . Thus, the equation can be rewritten after solving for $F(t)$ as:

$$F(t) = K \left[\frac{1 - e^{-(p+q)t}}{1 + \left(\frac{q}{p}\right)e^{-(p+q)t}} \right] \quad (2)$$

2.3.3. Logistic Model

The Logistic Growth model was introduced by Pierre Francois Verhulst around 1838, who used it for population studies (Lotfi et al., 2014). The basis of the model is that initially, the growth has a slow start, then it becomes approximately exponential. Later, the growth rate decreases linearly as it reaches the carrying capacity. According to Rietmann et al. (2020), several studies have also used the model to forecast technology diffusion over time accurately.

Furthermore, since the model's maximum and minimum parameters can be adjusted if the results significantly differ from reality, the logistic model is applicable to explaining long-term EV diffusion. The Logistic model can be expressed as follows:

$$F(t) = \frac{K}{1 + e^{-b_1(t-b_2)}} \quad (3)$$

Where $F(t)$ is the cumulative adoption at time t , K is the market potential, b_1 is the growth factor, e is the base of the natural logarithms, and b_2 is the time it takes for $F(t)$ to reach 0.5.

2.3.4. Gompertz Model

The Gompertz model was initially designed to describe human mortality and has been widely used in biology (Sato, 2021). Franses (1994) wrote that the model's

advantage is the ability to verify the stability of the saturation level. Additionally, the model has been applied in other areas, like new technology, where the adoption rate starts and ends slowly.

The Gompertz model describes growth that slows down as the population approaches a maximum limit, called the carrying capacity. The model assumes that the growth rate decreases exponentially as the population gets closer to the carrying capacity (Franses, 1994). In the context of Electric Vehicles (EVs), the Gompertz model has also been tested for its performance compared with other models (Kumar et al., 2022). The model can be written as:

$$F(t) = K \left[e^{-e^{-b_1(t-b_2)}} \right] \quad (4)$$

The notations used here are the same as previously defined. The Gompertz curve has two main characteristics: the growth rate is always positive, and it has a turning point before reaching its maximum point. Notably, while the model assumes the growth of the process is always positive, this may be different for a smaller data set. Thus, one solution is to remove any observations that do not support this assumption and then use the remaining data points to estimate the model.

2.3.5. Generalized Bass Model

While the Bass model is one of the most commonly used diffusion models, it has the drawback of not considering external factors affecting sales. Therefore, the Generalized Bass model was developed to address the drawback. For example, an additional variable, 'marketing mix,' is added to the existing Bass model. Thus, the Generalized Bass model can analyze the effects of incentives or promotions on innovations. The model can be expressed as follows:

$$\frac{dF(t)}{dt} = [p + qF(t)][1 - F(t)]x(t) \quad (5)$$

Where x is the percentage change in marketing mix variable, where: $x = 1 + \frac{\alpha[P(t)-P(t-1)]}{P(t-1)}$ and α is the coefficient representing the diffusion rate changes for every 1% change in $P(t)$. Thus, by increasing x , $\frac{dF(t)}{dt}$ is also increased, meaning that the marketing mix accelerates the spread of the innovation among the population. In this study, $P(t)$ will indicate the number of charging stations in each country.

2.4. Conclusion

In conclusion, several studies have identified factors that positively influence EV adoption, such as government incentives, consumer awareness, education, and the availability of charging infrastructure. Additionally, previous research has highlighted that diffusion models are a widely used tool in innovation adoption research. These models can be applied to predict EV uptake, considering the innovation and imitation effects on the spread of new technology. However, it is also essential to consider environmental, social, and economic factors to improve the accuracy of EV forecasting models.

3. METHODOLOGY

3.1. Data Collection

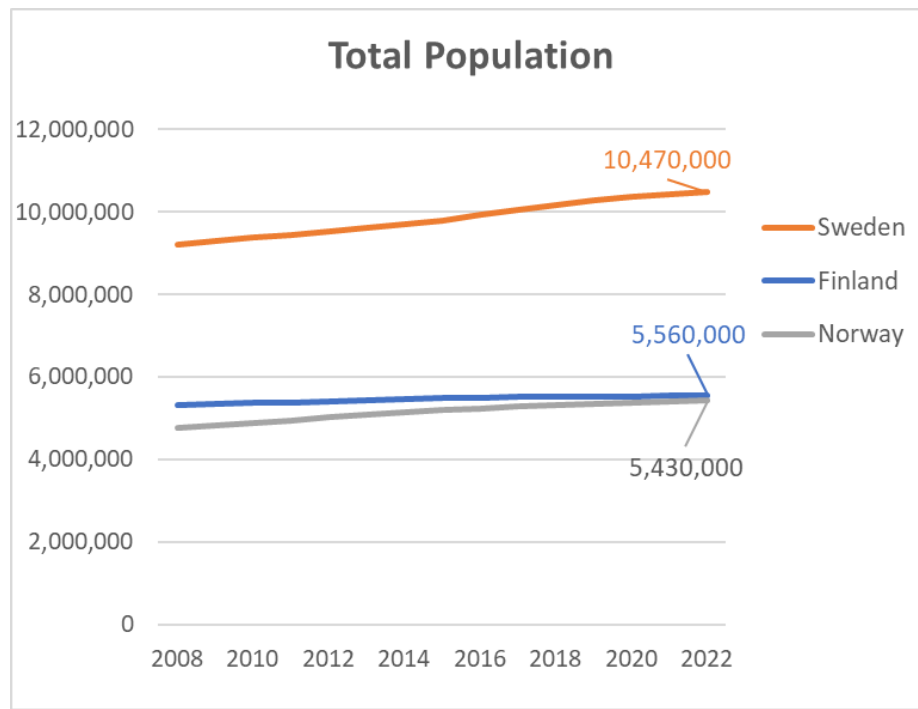


Fig 1. Total Population in Finland, Sweden, and Norway from 2008 to 2022 (WB, 2021)

To show comparisons between Finland, Sweden, and Norway, this paper needs to consider the differences in the countries' population size. The total population in Finland, Sweden, and Norway from 2008 to 2022 was extracted from the DataBank provided by the World Bank website (WB, 2021). Notably, according to Fig 1, Finland and Norway's population sizes were relatively similar from 2008 to 2022, while Sweden's population size was close to double Finland and Sweden in the same time period.

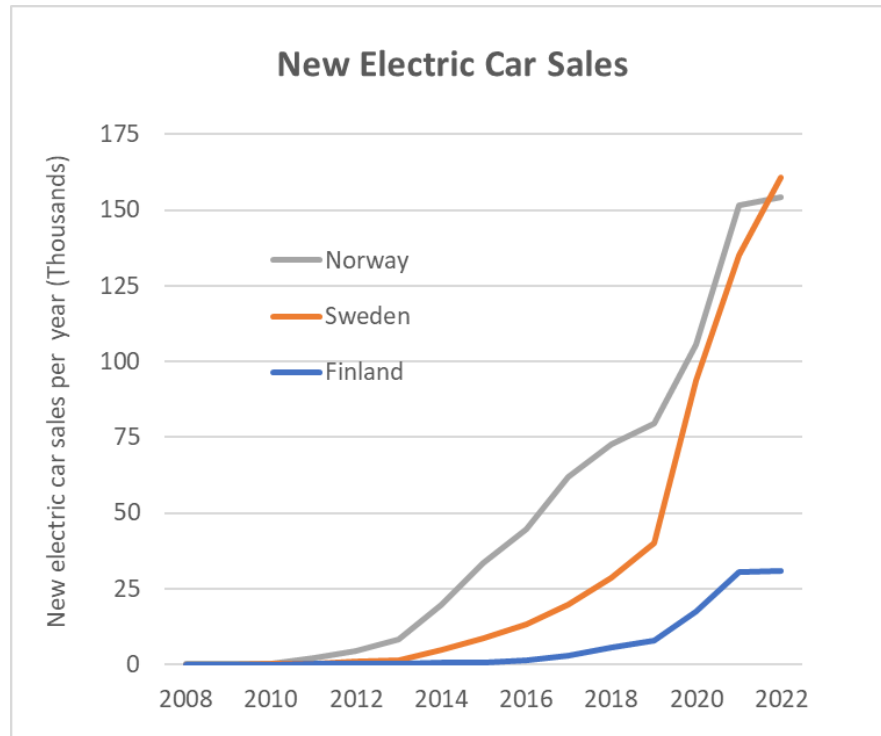


Fig 2. Electric car sales in Finland, Sweden, and Norway (EAFO, 2022)

The historical data about new registrations of electric passenger cars and electric buses in Finland, Sweden, and Norway are retrieved from the European Alternative Fuels Observatory website. The data includes battery electric vehicles (BEVs) and plug-in hybrid vehicles (PHEVs) covering from 2008 to 2022. Looking at Fig 2, it can be seen that Norway and Sweden's new electric car sales far exceeded Finland's.

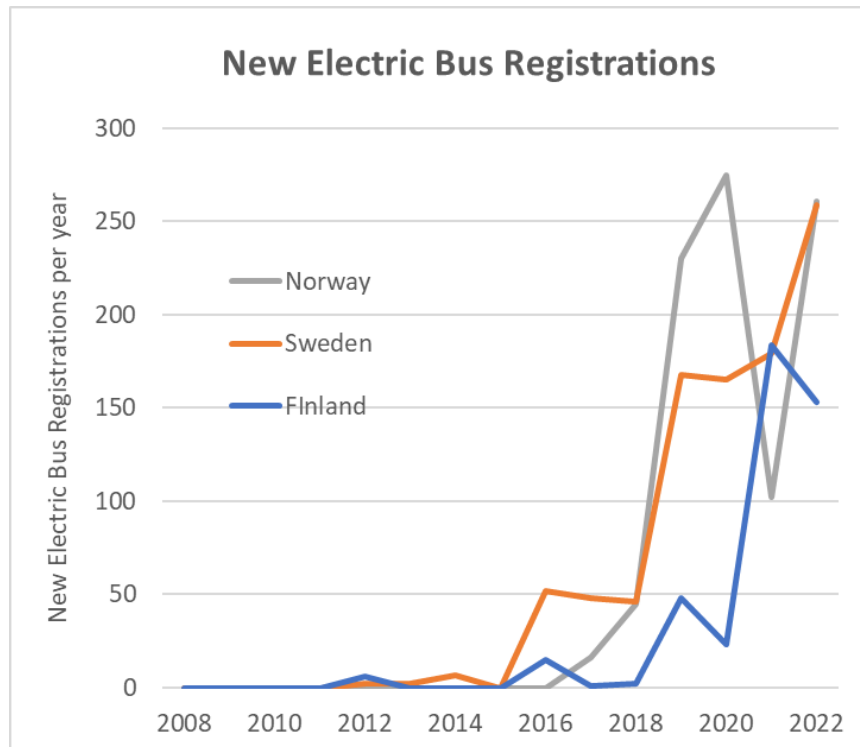


Fig 3. Electric bus registrations in Finland, Sweden, and Norway (EAFO, 2022)

Fig 3 shows the number of electric bus registrations in Finland, Sweden, and Norway. Noticeably, Norway's new electric bus registrations dropped dramatically in 2021. This could have been the effect of the COVID-19 pandemic reducing the demand for public transport. Furthermore, except in 2021, from 2016 to 2022, Norway and Sweden's numbers of new electric bus registrations were greater than Finland's.

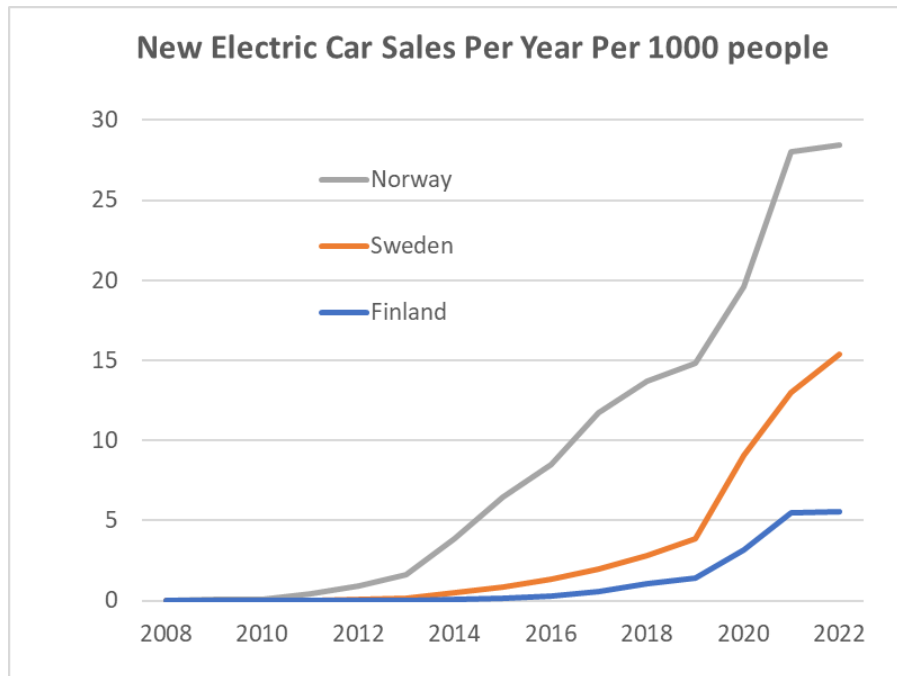


Fig 4. Electric car sales per 1000 people in Finland, Sweden, and Norway (EAFO, 2022)

To account for the different population sizes, Fig 4 shows Finland, Sweden, and Norway's new electric sales per 1000 people. In 2022, Norway had almost double the number of electric car sales per 1000 people compared to Sweden and nearly six times more than Finland.

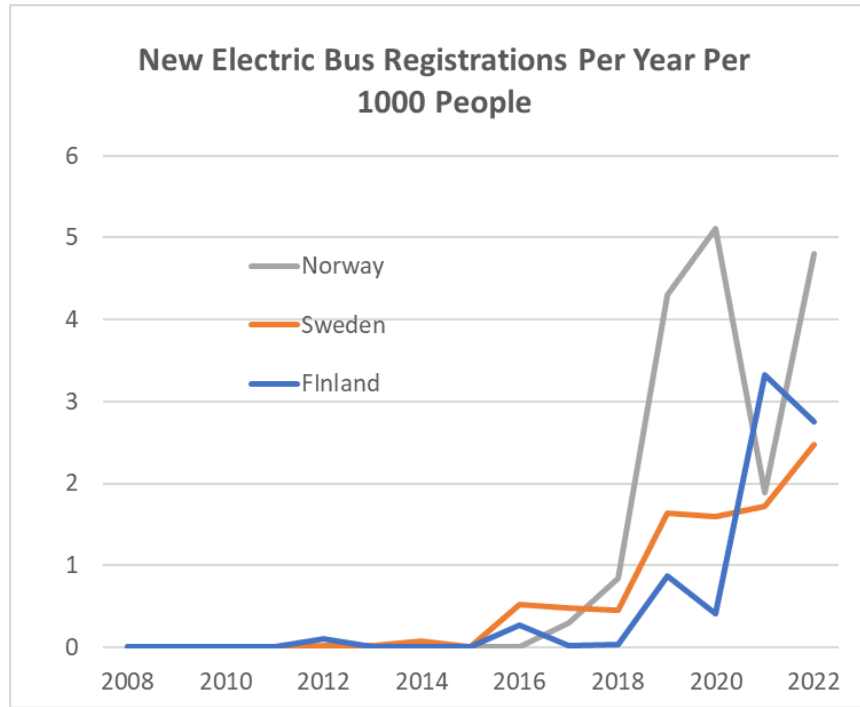


Fig 5. Electric bus registrations per 1000 people in Finland, Sweden, and Norway (EAFO, 2022)

Fig 5 illustrates that excluding 2021, Norway's electric bus registrations per 1000 people were more than Sweden's and Finland's from 2018 to 2023.

3.2. Research Methodology

This paper applies diffusion models to forecast new EV adoption up to 2030. The four models: Bass, Logistic, Gompertz, and Generalized Bass, will be fitted to the dataset. Then, the forecasted results will be used to calculate and show the cumulative adoption. The following details the methodology used in the research:

Step 1: Fitting the models to the historical data of new electric car registrations and the total number of electric buses for each country. Then the paper estimates the parameters for the diffusion models by setting the sum squared errors to a minimum value. The GRG Nonlinear method from the Excel tool 'Solver' is used to find the parameters.

Step 2: The paper will predict new registrations and the total number of electric cars and buses, including battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), until 2030. The time frame for the forecast is from 2023 to 2030. Finland, Sweden, and Norway are members of the EV30@30 initiatives, which aim to have 30% of new vehicle registrations be EVs by 2030 (IEA, 2017). Thus, the forecast will follow this scenario by assuming that the market potential for EVs in the three countries is 30% of the total vehicle market. Moreover, the market potential is similar for all four diffusion models.

According to Traficom (2022), in 2022, Finland had about 2.5 million cars and 7,500 buses. Meanwhile, Sweden had 5.9 million cars and 13,700 buses as of 2022 (transportstyrelsen.se). In addition, Norway had 3.7 million cars and 12,000 buses in 2022 (vegvesen.no). Therefore, the market potential for EVs in each country can be determined based on their current total market.

Step 3: Once the optimal parameter values are found, the paper chooses the best-fit diffusion models by comparing the Mean Percentage Error (MPE), Mean Absolute Error (MAD), and Mean Absolute Percentage Error (MAPE). These measures are determined by the following formulas:

$$\text{Mean percentage error} = \frac{100}{N} \sum_{j=1}^N \frac{x_j - \hat{x}_j}{\hat{x}_j} \quad (6)$$

$$\text{Mean absolute error} = \frac{\sum_{j=1}^N |x_j - \hat{x}_j|}{N} \quad (7)$$

$$\text{Mean absolute percentage error} = \frac{100}{N} \sum_{j=1}^N \frac{|x_j - \hat{x}_j|}{\hat{x}_j} \quad (8)$$

In the three formulas, x_j is the actual value for the new EV registrations in the j^{th} period, $\hat{x}_j$ is the predicted value for the new EV registrations in the same time frame. A lower MPE/MAE/MAPE value suggests a better model fit. Furthermore, a positive MPE indicates the model's forecast is overpredicting on average, while a negative MPE shows the model's forecast is underpredicting on average.

Step 4: The smallest MPE/MAE/MAPE model will be chosen as the best model for fitting the existing data. More weight is given to MAPE since it shows the true forecasting error making it easier to compare different models.

Step 5: Then, the forecasting results will be presented. This study also examines how the number of available charging points affects the diffusion process in each country. The Generalized Bass diffusion model considers the number of charging infrastructures when predicting the EV technology adoption rate. Finally, the paper will analyze the implications and provide conclusions about EV diffusion.

4. FINDINGS

4.1. Comparison Between the Diffusion Models

It is critical for researchers to assess the potential errors in forecasting models when predicting EV diffusion across countries. This paper calculates the four diffusion models to find the best-fit model and compares them based on their MPE, MAD, and MAPE values. Table 1 shows the MPE, MAD, and MAPE values as performance measures for forecasting, while Appendix Table 1 provides the optimal model parameters calculated in software Excel. The values formatted as bold in Table 1 indicate that they are closest to zero among the values in the four models. Accurately identifying the best model is crucial for predicting EV diffusion with the least errors.

Country	EV	MPE			
		Bass Model	Logistics Model	Gompertz Model	Generalized Bass Model
Finland	Car	8.849061966	7.905077503	3.530009795	8.986548289
	Bus	13.3657064	9.275113447	-30.50226648	15.14002713
Sweden	Car	6.410694153	5.5733291	-6.261191717	2.572424076
	Bus	-73.19743821	0.392816902	1.005428897	-69.60305464
Norway	Car	-3.14105038	-1.366639152	0.490180623	-6.738940337
	Bus	-1.196799613	27.96256115	17.88279624	-0.699214944
Country	EV	MAD			
		Bass Model	Logistics Model	Gompertz Model	Generalized Bass Model
Finland	Car	1671.729045	1581.74791	1024.14864	1998.507924
	Bus	8.652435949	8.506236846	8.338444825	8.717115583
Sweden	Car	4607.57896	4512.911467	4118.324592	4534.898736
	Bus	62.03294534	17.99984934	9.036822458	61.4875911
Norway	Car	7025.294391	5886.564562	8154.212172	7376.770984
	Bus	45.59622941	54.56440599	26.50260618	46.63931641
Country	EV	MAPE			
		Bass Model	Logistics Model	Gompertz Model	Generalized Bass Model
Finland	Car	22.83691837	21.19063767	15.75076046	25.57614547
	Bus	18.24507105	16.52874564	42.67352889	20.29791668
Sweden	Car	14.98272977	14.34700177	8.32306996	13.33820949
	Bus	77.90966013	6.731984995	4.020137623	74.22485783
Norway	Car	13.63378401	9.923188985	14.05579697	14.95051544
	Bus	41.96111205	33.69124848	29.99690215	42.60025886

Table 1. MPE, MAD, and MAPE values of the four diffusion models

Looking at MPE, the Gompertz performs the best for Finland's and Norway's electric car sales. Further, the MPE values for electric bus adoptions in Finland and Sweden for the Logistics model were the lowest. Sweden's electric car and Norway's electric bus diffusion have the lowest MPE value for the Generalized Bass model.

Moreover, with MAD, the Gompertz model best fits electric cars and buses in Finland and Sweden. The Gompertz model also has a good fit for electric buses in Norway. Meanwhile, the Logistics model has the lowest MAD for electric cars in Norway.

Using MAPE, the Gompertz model provides a good fit for electric cars in Finland and Norway. The Gompertz model also yields the lowest MAPE for electric buses in Sweden and Norway. Additionally, the Logistics model produces the lowest MAPE for electric cars in Norway and electric buses in Finland.

Overall, the Logistic and Gompertz models produce the best fit for EV adoption in Finland, Sweden, and Norway. Since more weight is given to MAPE, the two groups based on the diffusion models that produce the lowest MAPE values are as follows:

- Gompertz Model: Finland (electric cars), Sweden (electric cars and electric buses), and Norway (electric buses)
- Logistics Model: Finland (electric buses) and Norway (electric cars)

4.2. Forecast Results

The following figures show the comparison line graphs between actual and fitted diffusion predictions for new EV adoptions and the total number of EVs in Finland, Sweden, and Norway. The graph's x-axis represents the time from 2008 to 2030, while the y-axis displays the number of EVs for new adoptions per year and total adoptions. The historical data and the forecasted growth models are plotted in these figures.

4.2.1. Finland

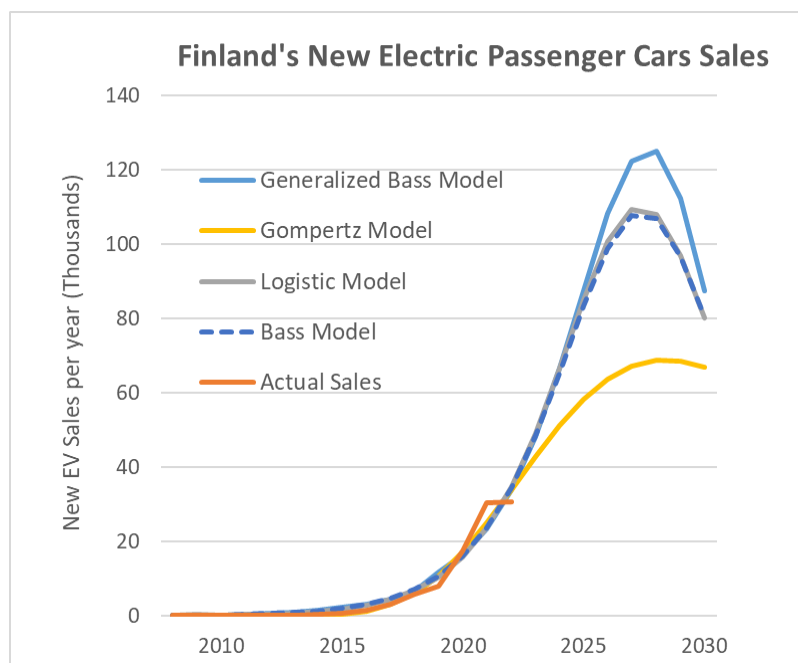


Fig 3. Forecast Finland's new electric car sales

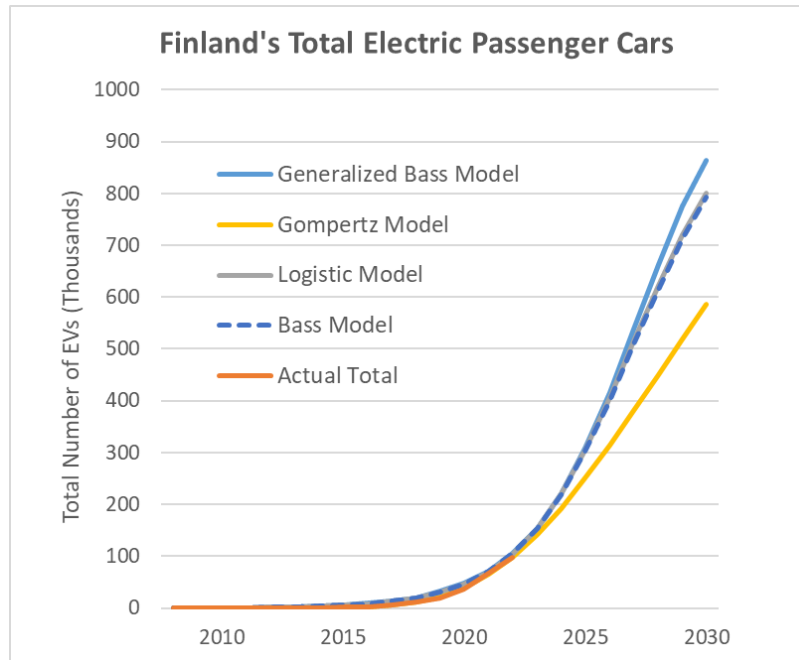


Fig 4. Forecast Finland's total number of electric cars

According to Fig 3, the Logistics and Bass models forecast that Finland's electric car sales will culminate at around 110,000 sales near 2028. Considering the increasing number of charging stations, the Generalized Bass model predicts that there will be more than 120,000 sales in 2028. Meanwhile, the Gompertz model forecasts 70,000 sales in 2028.

Compared to the Logistics model, the Gompertz model assumes a faster decrease in the growth rate as the total adoption approaches the market potential (Franses, 2019). Thus, before reaching the market potential, the Gompertz model has a slower adoption rate than the Logistics model. However, comparing the growth rates predicted by the Bass, Generalized Bass, and Gompertz models would not be meaningful since the three models are based on different assumptions and are used to describe different processes (Satoh, 2021). Therefore, this study will not compare the adoption rate between the models. Instead, it will see which diffusion process best fits the current EV sales.

Based on Fig 4, the Generalized Bass, Logistics, and Bass models forecast that Finland's total number of electric cars will be around 820,000. The Gompertz model forecasts the total to be almost 600,000. The results align with the goal of having 700,000 electric cars in Finland in 2030, set by the Finnish Ministry of Transport and Communications (Yle, 2022).

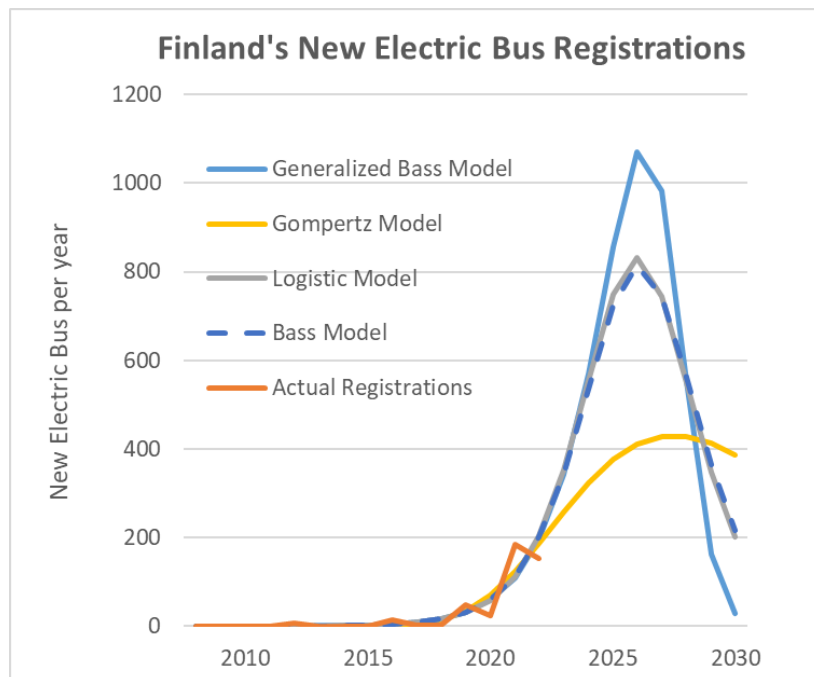


Fig 5. Forecast Finland's new electric bus registrations

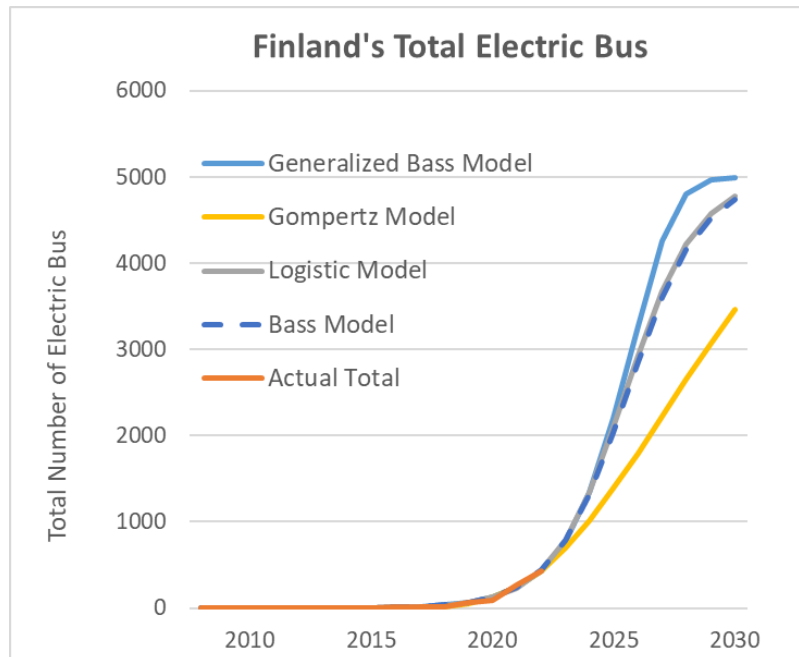


Fig 6. Forecast Finland's total number of electric buses

Regarding electric buses, Fig 5 illustrates that new electric bus registrations will peak at around 800 in 2026 based on the Bass and Logistics models. The Generalized Bass model forecasts that electric bus registrations will reach over 1000 in 2026. In addition, the Gompertz model predicts a less dramatic rise and drop in the electric bus adoption process, reaching a peak of over 400 in 2028.

Concerning Fig 6, the Generalized Bass, Logistics, and Bass models forecast the total number of electric buses to be around 5000 in 2030. On the other hand, the Gompertz model shows that the total number of electric buses will be 3500 in 2030.

4.2.2. Sweden

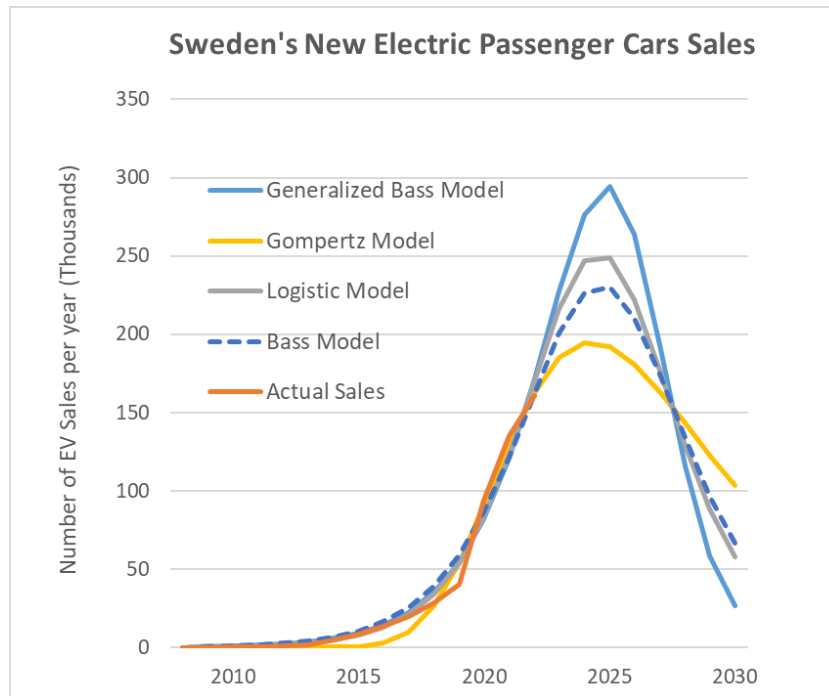


Fig 7. Forecast Sweden's new electric car sales

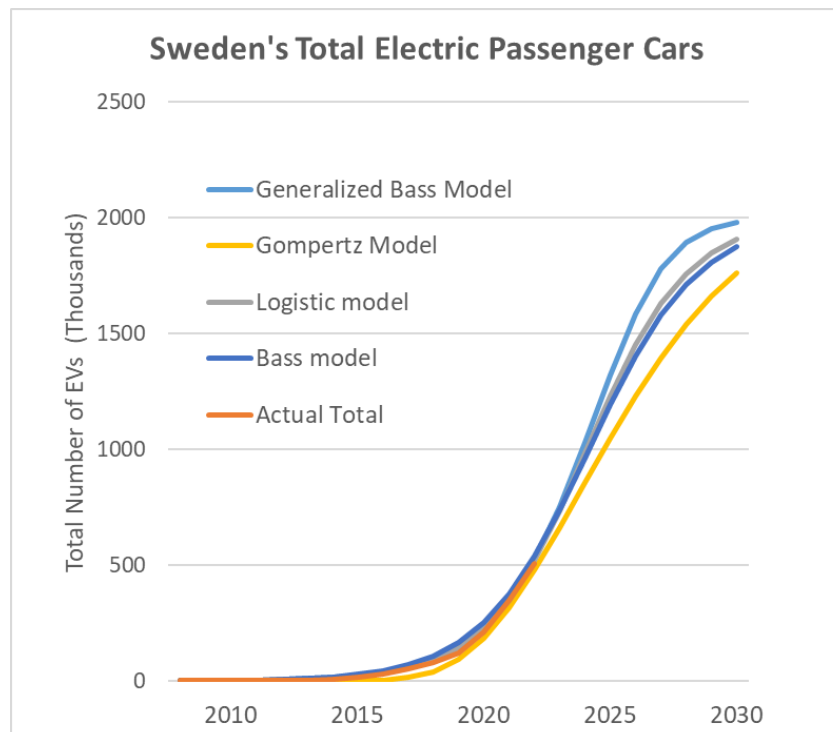


Fig 8. Forecast Sweden's total number of electric cars

Corresponding to Fig 7, the Generalized Bass, Logistics, Bass, and Gompertz models forecast EV sales in Sweden to be highest in 2025, reaching 300,000, 250,000, 230,000, and 200,000, respectively.

As for Fig 8, the four models forecast the total number of electric cars in Sweden to range between 1,750,000 and 2,000,000 by 2030. It is worth noting that in 2022, there were 288,000 all types of cars sold and 5,700,000 cars currently in use in Sweden, according to the ACEA (2022) report.

The results of the models suggest that the Swedish government's targets of achieving 80% electric car sales by 2030 and banning the sale of petrol and diesel vehicles after 2030 could be realized if EV adoption follows the diffusion processes described in any of the four models.

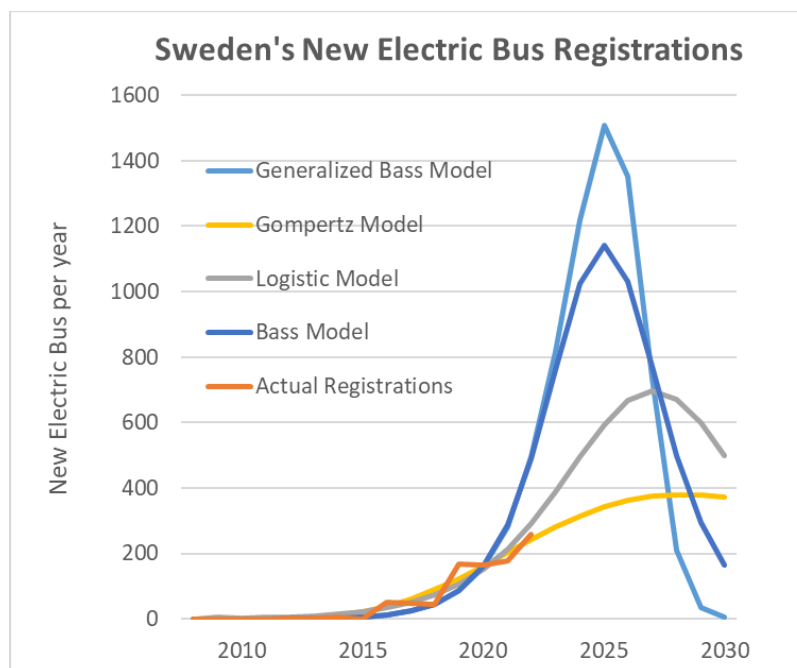


Fig 9. Forecast Sweden's new electric bus registrations

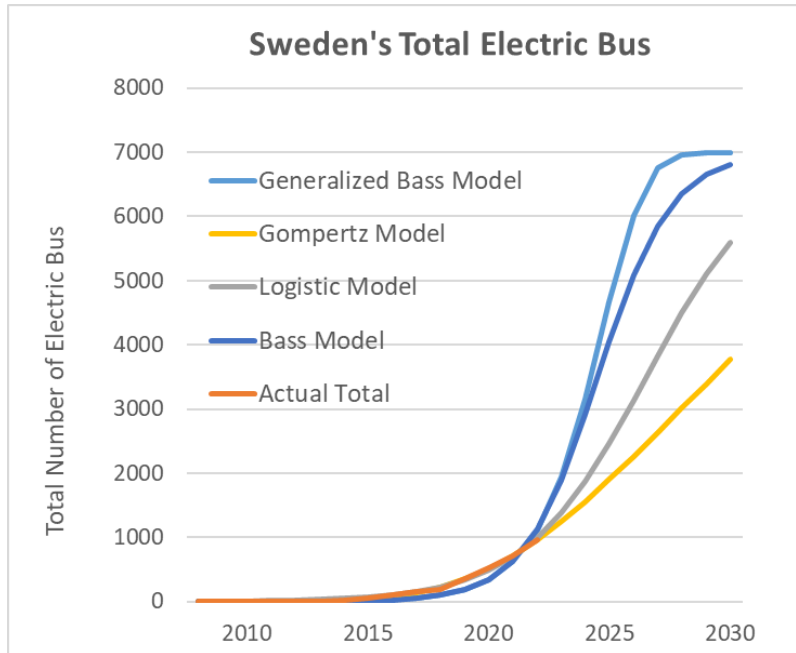


Fig 10. Forecast Sweden's total number of electric buses

Fig 9 shows many differences in the adoption process of Sweden's electric bus between the four models. The Generalized Bass and Bass models predict that the number of electric bus registrations will increase to 1500 and 1100 in 2025, respectively. Further, the Logistics and Gompertz models forecast that 700 and 400 new electric buses will be registered in 2027.

Regarding the total, in Fig 10, the Generalized Bass and Bass models have the total number of electric buses in Sweden to be 7000 in 2030. By 2030, the Logistics model estimates there will be 5500 electric buses, and the Gompertz model speculates almost 4000 electric buses.

4.2.3. Norway

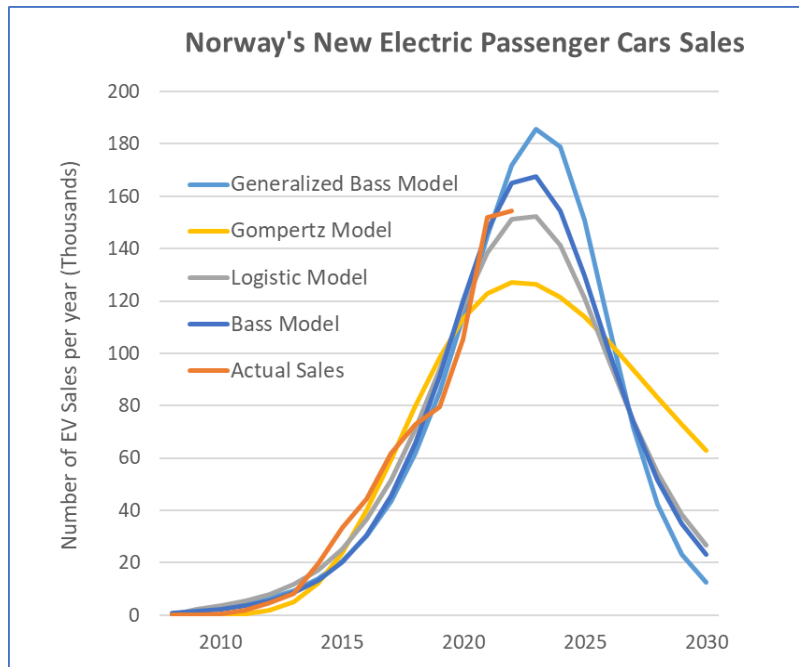


Fig 11. Forecast Norway's new electric car sales

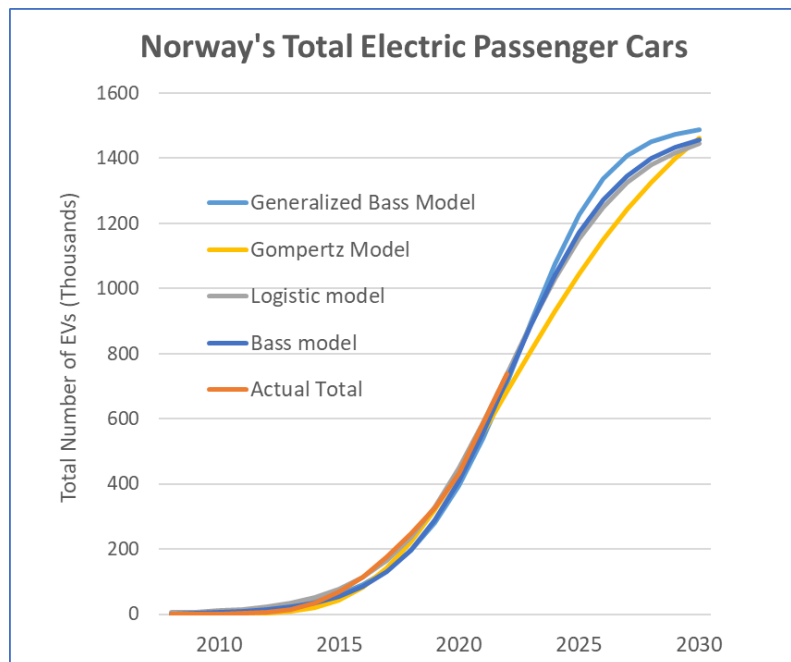


Fig 12. Forecast Norway's total number of electric cars

Corresponding to Fig 11, the Generalized Bass, Logistics, Bass, and Gompertz models predicted the highest number of electric car sales in Norway in 2023. The corresponding sales figures predicted by each model are 185,000, 150,000, 170,000, and 130,000, respectively. The forecast results align with the Norwegian Parliament's plan for all new cars sold to be electric by 2025 (elbil.no)

According to Fig 12, the four models forecast the total number of electric cars in Norway to be over 1,400,000 by 2030.

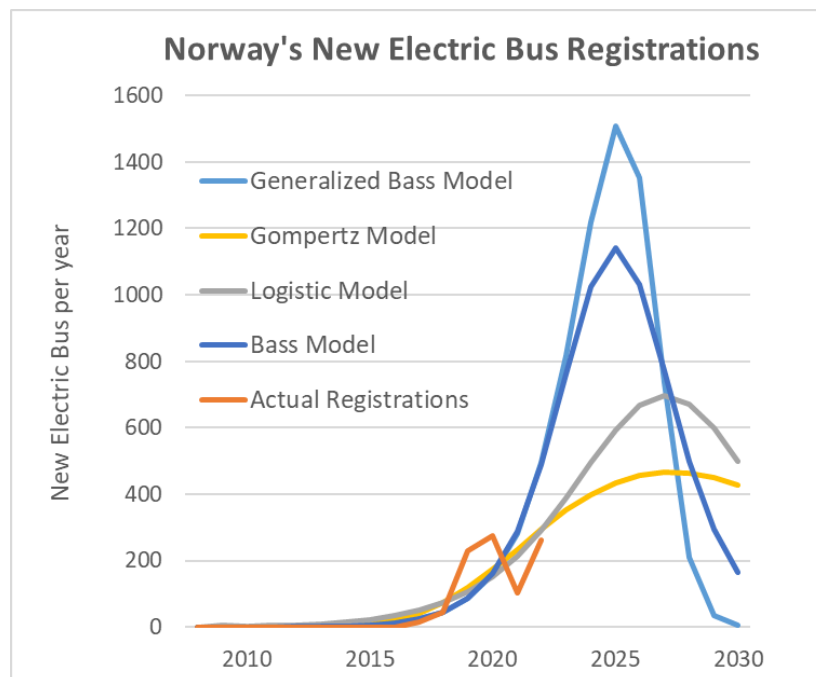


Fig 13. Forecast Norway's new electric bus registrations

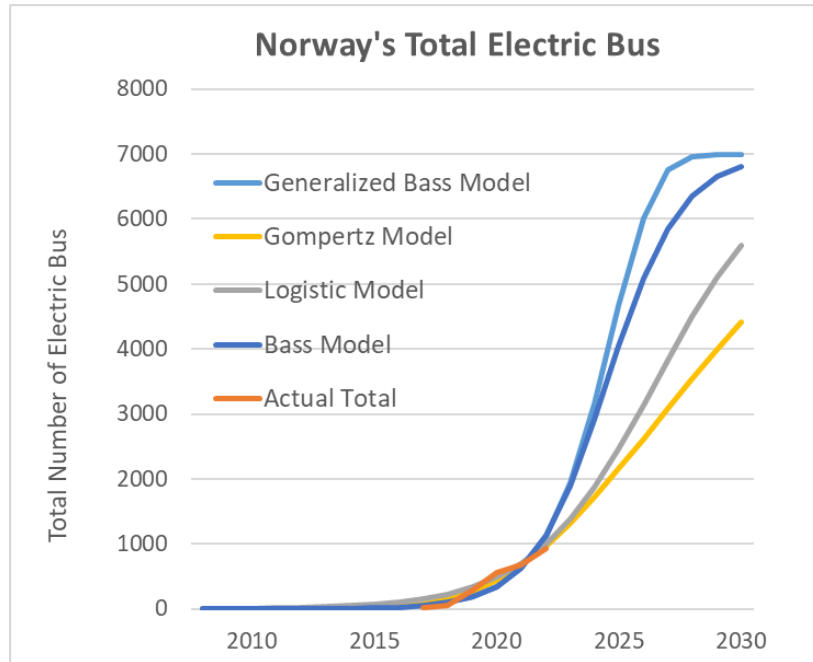


Fig 14. Forecast Norway's total number of electric buses

Fig 13 shows that, based on the Generalized Bass and Bass models, there will be a peak in new electric bus registrations at 1500 and 1100 in 2025. The Logistics model predicts that electric bus registrations will be 700 in 2027. Meanwhile, the Gompertz model predicts a less pronounced increase and decrease in the adoption of electric buses, with a peak of 480 in 2027.

As for Fig 6, the Generalized Bass and Bass models project that the total number of electric buses will be 7000 by 2030. In addition, the Logistics model shows there will be 5500 electric buses in 2030. Finally, the Gompertz model suggests that the total number of electric buses will be 4400 by 2030.

It is important to note that the Bass model did not provide the best fit for EV diffusion based on the lowest MPE, MAD, and MAPE among Finland, Sweden, and Norway. This is somewhat surprising since some previous studies on forecasting EV sales have relied on the Bass model (Kumar et al., 2021). For example, Wu et al. (2019) used the Bass model to forecast EV sales in China. Lee et al. (2019) also forecast EV market diffusion in California using the Bass model. Therefore, it is reasonable to assume that different markets have different EV diffusion processes. Thus, this

paper suggests using different diffusion models depending on each country's MPE, MAD, and MAPE values.

5. DISCUSSION AND ANALYSIS

5.1. Influence of Charging Points

This study utilizes the Generalized Bass diffusion model to examine the impact of charging infrastructure on EV adoption rates. The model incorporates external factors into diffusion forecasting. In this case, charging points are considered an external variable that can accelerate EV diffusion across Finland, Sweden, and Norway. While Soumia and Maaroufi (2018) have previously used the Generalized Bass diffusion model to investigate the relationship between cumulative EV sales and battery prices in Morocco, there is a lack of studies considering the number of charging points in their diffusion models. Thus, this study explores this relationship using historical data for each country. Nonetheless, several studies have confirmed that insufficient charging stations are a major obstacle to EV diffusion (Noel et al., 2020).

According to EAFO (European Alternative Fuels Observatory) (2022), the number of EV charging points in Finland, Sweden, and Norway reached a total of 6121, 25,465, and 26,178, respectively, in 2022. Thus, in 2022, Finland had 1 charger per 25 EVs, and Sweden had a ratio of 1 charger per 20 EVs. Interestingly, Norway had a relatively lower number of chargers in 2022, with 1 charger per 30 EVs. Compared to the relatively constant charging point ratio of 1 to 10 EVs in China, Korea, and the Netherlands from 2015 to 2021, Finland, Sweden, and Norway's charging point ratio decreased in the same period (IEA, 2022). This reflects the charging point installations being outstripped by EV adoptions in the three Nordic countries.

The markets in Finland, Sweden, and Norway are distinguished by their reliance on home charging due to the large number of single-family dwellings having garages compared to global standards (IEA, 2022). Countries that depend more on public charging seem to be expanding their charging networks as required. On the other hand, countries with a significant proportion of residential charging may not require as many public charging points to serve more electric vehicles. However, as more people

shift from conventional vehicles to electric ones, even countries with many single-family homes will need to rely more on public charging options.

Accessible public charging stations are critical for the successful deployment of electric vehicles. Thus, it is essential to understand the diffusion models that consider the development of charging infrastructure in their forecast. In addition, the Generalized Bass models in Section 4.2. illustrate that the external factor of charging points continuously supports EV growth until the EV adoption rate peaks and declines by 2030, 2027, and 2026 for Finland, Sweden, and Norway. As a result, this suggests that these three countries will attain their EV30@30 initiatives, an EV market share of 30% before 2030.

These countries have implemented various policies to develop their EV charging infrastructure. Some significant measures driving the growth of private charging facilities include cost-effectiveness, convenience, purchasing incentives, rebates, and preferred rates (IEA, 2022).

5.2. Effects of COVID-19 on EV diffusion

5.2.1. Electric Passenger Cars

The COVID-19 pandemic has significantly impacted Finland, Norway, and Sweden's EV markets. While the pandemic initially caused a slowdown in EV sales, the markets have shown resilience and have continued to grow despite the challenges posed by the pandemic (IEA, 2021).

According to the Finnish Transport and Communications Agency (Traficom) (2021), the sales of new EVs in Finland, including both battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), increased significantly in 2020 and 2021. In 2020, a total of 17,475 new electric vehicles (BEVs and PHEVs) were registered in Finland, which was a 122% increase compared to the previous year. In 2021, EV sales increased by 74% compared to 2020.

The Swedish Transport Agency (2021) has also reported a substantial increase in new EV sales in Sweden during 2020 and 2021. In 2020, Sweden registered

93,728 EVs, 134% more than in 2019. Moreover, in 2021, Sweden had 44% more EV sales than the previous year.

Similarly, EV sales in Norway continued to increase during the COVID-19 pandemic. The EV registrations in 2020 were 33% greater than in 2019. Additionally, the number of registrations in 2021 was 44% greater than in 2020.

The impact of COVID-19 on electric car diffusion in Finland, Sweden, and Norway has been relatively minor compared to other countries, thanks to strong government support for EVs and a robust EV market. The pandemic has also highlighted the importance of sustainable transportation and the role of EVs in achieving long-term environmental goals.

One possible explanation for the growth of the EV market during the pandemic is the increased focus on sustainability and environmental protection (IEA, 2021). As the pandemic brought about temporary improvements in air quality and reduced traffic congestion, many consumers and policymakers began to prioritize the transition to cleaner, more sustainable modes of transportation.

Furthermore, the EV30@30 initiative has prompted Finland, Sweden, and Norway to implement measures to mitigate the impact of COVID-19. Table 2 summarizes a few of the measures taken by these countries to support EV diffusion during the COVID-19 pandemic.

Despite the initial slowdown caused by the pandemic, the electric vehicle market in Finland, Norway, and Sweden has shown remarkable growth in 2020 and 2021. The market resilience suggests that EV adoption will continue growing in these countries.

Country	Policies description
Finland	1. Zero-emission passenger cars are exempt from registration tax 2. Purchase incentive of €2,000 granted for households to purchase or lease a new BEV in class M1 (passenger car) provided value ≤ €50,000. 3. Users will be able to charge BEVs and PHEVs at the workplace and at public charging points tax-free.
Sweden	1. Low annual road tax (SEK 360 ~ €32) for zero-emission vehicles and PHEVs. 2. Climate bonus for light vehicles: SEK 70,000 (€6,260) for BEVs, SEK 44,417 (€4023) for PHEVs with 1g CO₂/km down to SEK 10,020 (€895) for ≤ 60g CO₂/km. 3. Between 2018 and 2020, SEK 90 million (EUR 8,7 million) will be annually allocated to support home chargers with up to SEK 10,000 (EUR 960) for hardware and installation costs.
Norway	1. Electric cars are exempt from the registration fee as of 2021. 2. EVs are granted a reduction and only pay a scrapping fee, EUR 249. NOK 455 (EUR 48) is the annual road tax for both fully electric vehicles and plug-in hybrids. 3. Until the end of 2021, there was no CO₂ tax, no weight tax, and no VAT benefit for purchasing EVs. 4. The Norwegian Electric Car Association (Elbilforeningens) gives users access to charging units all over the country at a reduced price.

Table 2. Policy measures to support EV diffusion during COVID-19 (EAFO, 2022)

5.2.2. Electric Public Bus

In the three Nordic countries, the adoption of electric buses has been steadily increasing in recent years as a part of efforts to reduce greenhouse gas emissions and improve air quality. Nevertheless, the pandemic has caused disruptions to this progress in some ways.

In Finland, the number of newly registered e-buses decreased in 2020 due to the pandemic. There were 28 newly registered e-buses in 2020, a 42% decrease from 2019 (traficom.fi). This decline can be attributed to reduced public transport services during the pandemic and the resulting budget cuts. However, electric bus registrations recovered to 184 in 2021.

Sweden's electric bus adoption process was less affected by the pandemic. According to the Swedish Transport Agency, the number of new electric buses in 2020 was 165, a decrease of 1.8% compared to 2019. Then, the registrations increased to 179 in 2021.

Interestingly, the timing of the pandemic's impacts on electric bus registrations in Norway was different. In 2020, Norway's new electric bus adoptions were 275, 19.6% more than in 2019. Electric bus registrations in Norway increased in 2020 due to a combination of factors such as strong policy support, incentives for electric vehicle adoption, and increased awareness of the benefits of electric buses (Reuter, 2021). However, new registrations dropped by 62.9% the following year, resulting in only 102 new electric buses in 2021 before returning to 261 in 2022. A few possible reasons for this decrease are supply chain disruptions and delays in the delivery of buses (electrive, 2022).

Overall, the COVID-19 pandemic impacted electric bus registrations in Finland, Sweden, and Norway. While the pandemic caused a decrease in electric bus adoptions in the three countries, the adoptions were back to normal in 2022. As the countries recover from the pandemic, the electric bus market has shown some resilience and is expected to grow.

6. CONCLUSIONS

6.1. Main Findings

To accurately predict the future scenarios for EV adoption in Finland, Sweden, and Norway, it is crucial to have a deep understanding of the EV diffusion process with as few errors as possible. This study utilized four diffusion models, including Bass, Gompertz, Logistic, and Generalized Bass, to forecast the future of EV technology in the three Nordic countries. The findings of this study reveal variations in the diffusion process across Finland, Sweden, and Norway regarding the diffusion rate. This research has gained insights related to the following research questions.

Which models would best forecast the electric car and electric bus demands in Finland, Sweden, and Norway?

This paper evaluated the forecast results of all four diffusion models and examined the most appropriate model based on the MPE, MAPE, and MAD values. When looking at MPE, the Gompertz model best predicted electric car sales in Finland and Norway, while the Logistics model performed best for electric bus adoptions in Finland and Sweden. The Generalized Bass model best fits the diffusion of electric cars in Sweden and electric buses in Norway.

Based on MAPE, Finland's electric cars, Sweden's electric cars and buses, and Norway's electric buses were best predicted by the Gompertz model. Further, the Logistics model best predicted Finland's electric buses and Norway's electric cars. Regarding MAD, the Gompertz model best fits electric cars and buses in Finland and Sweden and electric buses in Norway. Moreover, the Logistics model had the lowest MAD for electric cars in Norway.

In general, considering all three measures with more weight given to MAPE, the Logistic and Gompertz models would be preferable for forecasting EV markets in Finland, Sweden, and Norway.

How do charging points affect EV diffusion in Finland, Sweden, and Norway?

This study examined the impact of charging points on the EV diffusion process and projected future charger requirements in Finland, Sweden, and Norway until 2030. In Finland, Sweden, and Norway, the Generalized Bass model, which incorporated charging points in its forecasts, predicted EV adoptions to decline after peaking before 2030, indicating a 30% EV market share by 2030. The study also highlights the relationship between the number of EVs and the number of charging stations. As of 2022, Finland, Sweden, and Norway had fewer charging points per EV than the global average (IEA, 2022). This is due to home charging being widely available in these countries. However, as EVs become more accessible, the three Nordic countries should rely more on public charging options.

What are the effects of the COVID-19 pandemic on electric car and electric bus diffusions in Finland, Sweden, and Norway?

Regarding electric cars, the COVID-19 pandemic initially slowed down EV sales in Finland, Norway, and Sweden. However, the markets showed resilience and continued to grow despite the challenges posed by the pandemic. The impact of COVID-19 on the EV market in these countries has been relatively minor due to strong government support and a robust market. The pandemic has emphasized the importance of sustainable transportation and the role of EVs in achieving long-term environmental goals. The growth of the EV market during the pandemic can be attributed to the increased focus on sustainability and environmental protection. The EV30@30 initiative has prompted measures to support EV diffusion during the pandemic in these countries. Overall, the electric vehicle market in Finland, Norway, and Sweden is expected to continue growing.

On the other hand, the pandemic disrupted electric bus adoptions in Finland, Sweden, and Norway in 2020. The disruptions were due to reduced public transport services during the pandemic, supply chain issues, and delivery delays. However, in 2022, the electric bus market has recovered and is expected to grow as the countries transition from the pandemic.

6.2. Implications for International Business

The use of diffusion models to predict the adoption of electric cars and buses in Finland, Sweden, and Norway has significant implications for businesses operating internationally. Firstly, this study provides valuable insights for automobile and bus manufacturers seeking to expand into the electric vehicle markets of these countries. With a better understanding of the potential growth rates and adoption patterns, companies can allocate their resources more effectively, meet demand, and take advantage of emerging opportunities.

Secondly, the findings of this research underline the crucial role of government policies and incentives in driving electric vehicle adoption. International businesses must be cognizant of the regulatory landscape in each country and tailor their

strategies accordingly. For instance, variations in tax incentives or subsidies for electric vehicle purchases may influence consumer behavior, ultimately impacting market demand. Furthermore, forecasting EV markets can assist governments in anticipating future energy requirements with more emphasis on renewable and clean energy sources.

Thirdly, the diffusion models employed in this study are transferable to other markets and industries, providing insights that can aid businesses operating across different regions. By analyzing adoption patterns and identifying key drivers of diffusion, companies can better gauge market potential and tailor their strategies to maximize their success.

In conclusion, the use of diffusion models to forecast the uptake of electric vehicles is vital for international businesses, enabling them to make strategic decisions, comprehend government policies, anticipate future energy demand, and adapt to changing market trends. As the world moves towards sustainable transportation, these insights will be invaluable for companies seeking to navigate the dynamic landscape.

6.3. Limitations

The application of diffusion models, such as the Bass, Gompertz, Logistics, and Generalized Bass models, in forecasting EV adoptions has provided some insights. Nevertheless, it is essential to recognize that this study still has limitations.

To begin with, the accuracy of these models depends on the quality of the data used for fitting. Any inaccuracies or biases in the data may affect the predictions made by the models. Additionally, the Bass, Logistics, and Gompertz models are based on the assumption that the factors driving adoption remain unchanged over time. However, in reality, these factors may fluctuate, making it challenging to forecast future adoption rates accurately. Although the Generalized Bass model accounts for external factors that could impact EV diffusion, its accuracy relies heavily on assumptions about the external factors.

Furthermore, these models are designed to capture the aggregate behavior of a population and may not reflect individual differences in adoption patterns. As such, using these models to predict the adoption rates of specific EV models or brands can provide mixed results.

Moreover, it is worth noting that this study solely focuses on the Nordic countries of Finland, Sweden, and Norway. Therefore, the findings may not be generalized to regions or countries with different cultural, economic, or political backgrounds.

Additionally, while this study only uses the MPE, MAE, and MAPE to determine the models' predictability, there are alternative methods to evaluate the predictive performance of these models. For instance, metrics such as F statistics, Pseudo R-squared, Akaike information criterion, etc. are available in different software packages can also be used in future studies.

All in all, while this research has provided valuable insights into EV diffusion, these limitations must be acknowledged. Future research should address these constraints to provide more accurate predictions and better comprehend the factors driving electric vehicle adoption.

6.4. Suggestions for Further Research

The use of diffusion models to forecast electric cars and buses in Finland, Sweden, and Norway has provided valuable insights into EV future demands. However, there are still several areas that require further investigation. This section offers suggestions for future research that could build on the findings of this thesis and contribute to a deeper understanding of EV diffusion.

Future research can explore the impact of socioeconomic and environmental factors on the diffusion of electric vehicles in other markets. Analyzing the effects of government policies, incentives, infrastructure, consumer attitudes, and income levels on electric car and bus demands in different markets would be valuable for governments, automakers, and utility providers. A comparative study of these factors

across different regions could provide more insights into the factors influencing the EV markets.

Another possible research area is to examine the applicability and accuracy of other diffusion models not used in this study. While the Bass, Logistics, Gompertz, and Generalized Bass models have provided useful forecasts, other models available could offer different insights into the diffusion of electric vehicles. For instance, the Weibull, Sigmoid, and Generalized Logistic models are widely used in the literature and could be applied to study how EV technology diffuses.

In addition, future research could investigate the impact of market uncertainties on EV adoptions in other countries. Market uncertainties, such as a pandemic, can cause significant changes in consumer behaviors and government policies, affecting electric vehicle demand.

In conclusion, this thesis has demonstrated the usefulness of diffusion models in forecasting the adoption of electric vehicles in Finland, Sweden, and Norway. However, several areas still require further investigation, including the impact of socioeconomic and environmental factors, the applicability of alternative diffusion models, and the impact of market uncertainties on different regions and markets.

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APPENDICES

Models			Bass		Logistics	
Country	EV	K	p	q	b1	b2
Finland	Car	1000000	0.000061	0.434298	0.442376	19.863268
	Bus	5000	0.000003	0.652312	0.672538	18.485435
Sweden	Car	2000000	0.000131	0.462666	0.506532	17.069003
	Bus	7000	0.000005	0.652312	0.398688	19.526836
Norway	Car	1500000	0.000393	0.448984	0.410788	15.075424
	Bus	7000	0.000005	0.672312	0.376877	18.483622

Models			Gompertz		Generalized Bass		
Country	EV	K	b1	b2	p	q	α
Finland	Car	1000000	0.156221	19.875234	0.000118	0.497004	0.0999999507
	Bus	5000	0.195202	18.950058	0.000012	0.869430	0.0000000057
Sweden	Car	2000000	0.241520	15.753881	0.000286	0.588404	0.0595367790
	Bus	7000	0.129374	19.808721	0.000024	0.869430	0.0000000045
Norway	Car	1500000	0.206300	13.794388	0.001288	0.493674	0.0494672934
	Bus	7000	0.158435	18.711657	0.000028	0.919430	0.0000000053

Table A1. Optimal values of model parameters